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Cross-Selling in Bank-Household Relationships: Mechanisms and Implications for Pricing

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1 Big picture

- **Question.** Do banks cross-sell loans and future deposits to existing household depositors, what mechanism drives this cross-selling, and do banks price deposits today in anticipation of future relationship profits?
- **Setting.** The paper uses Norwegian administrative relationship-level data covering all household-bank deposits and loans from 2004 to 2018. This is unusual because most banking studies observe either the loan side or the deposit side, not both sides of the same household-bank relationship.
- **Main fact 1: deposit stickiness.** A household observed with deposits at bank b is about 90 percentage points more likely one year later to remain at that bank than to deposit at an otherwise available bank. This persistence declines only slowly over the following 14 years.
- **Main fact 2: depositor-borrower conversion.** Existing depositors are about 20 percentage points more likely to later borrow from their deposit bank than comparable households in the same municipality, age, salary, and education groups with deposits elsewhere.
- **Mechanism.** Existing depositors pay a loan premium rather than receiving a discount, and banks do not appear better at identifying future defaulters among existing depositors. This points toward demand complementarities, not supply-side informational advantages.
- **Demand channel.** The evidence is more consistent with switching costs and household inaction than with stable unobserved preferences for particular banks. Mover and tenure tests are central for this conclusion.
- **Pricing implication.** Banks internalize future lending value when setting deposit prices. Relationships predicted to have higher future loan propensity receive better deposit terms, and their deposit spreads are less sensitive to policy rates at low-rate times.
- **Takeaway.** Deposits are not only cheap funding. They are also an onboarding product that gives banks a future option to sell mortgages and other loans to sticky household customers.

2 Incremental contribution

- **Relationship-level asset-liability link.** The paper connects the liability side and asset side of household banking using the same household-bank relationship, rather than inferring cross-selling from bank-level aggregates.
- **Direct measurement of cross-selling.** The authors observe when a deposit relationship precedes a loan relationship with the same bank, allowing direct measurement of depositor-borrower conversion over a 14-year horizon.
- **Demand versus supply mechanisms.** The paper separates demand complementarities from supply complementarities by testing loan pricing, future default screening, mover subsamples, and tenure effects.
- **Household banking contrast with firm banking.** The paper shows a mechanism different from much relationship-lending evidence for firms: for households, mortgage collateral makes informational advantages less important, while switching costs and inertia are central.
- **Pricing with customer lifetime value.** The paper shows banks internalize future cross-selling value in current deposit rates, connecting retail banking to marketing-style customer lifetime value.
- **Deposit-rate pass-through.** By linking cross-selling potential to the deposit spread beta, the paper adds a new relationship-level mechanism for why deposit rates may become less sensitive to policy rates.
- **Policy relevance.** The results speak to bank market power, household inertia, deposit competition, monetary transmission, and the value of banking franchises.

3 Data and measurement

3.1 Relationship-level banking data

- The main data come from Norwegian tax-register information on every deposit and loan relationship between individuals and Norwegian banks.
- The data report deposit and loan amounts and the interest paid or received in each relationship year.
- Annual data cover 2004–2018, providing a long window for tracking whether deposit accounts convert into loan relationships.
- Most Norwegian household loans are mortgages, and more than 99% of deposits have maturity below one year, making the setting especially relevant for deposit pricing and mortgage cross-selling.

3.2 Household, bank, and municipality variables

- Household controls include age group, salary group, education group, taxable real estate, debt, municipality, and number of bank relationships.
- Bank controls include bank-level balance-sheet characteristics and market shares.
- Municipal deposit and loan market concentration are measured with HHIs, and bank-level concentration variables are exposure-weighted across municipalities.
- The paper creates hypothetical household-bank relationships with all banks present in the household’s municipality, allowing actual relationships to be compared with plausible alternatives.

3.3 Key variables

- **Deposit spread:** policy rate minus deposit rate. A high spread means the depositor receives a relatively low deposit rate.
- **Loan spread:** loan rate minus policy rate. A high spread means the borrower pays a high loan rate relative to the policy rate.
- **Deposit interest lost:** the gap between the relationship’s deposit rate and the highest deposit rate available in the same municipality-year, multiplied by the relationship’s deposits.
- **Depositor-borrower conversion:** the event that a household later borrows from a bank where it had a prior deposit relationship.
- **Predicted loan propensity:** predicted probability that a relationship will become a borrower relationship; used to test deposit pricing.

4 Models and empirical specifications

4.1 Deposit persistence

$$I(\text{Depositor})_{hbt+\tau} = \alpha + \beta I(\text{Existing Depositor})_{hbt} + \delta_{mt} + \varepsilon_{hbt}.$$

Interpretation: The coefficient β measures how much more likely a household-bank relationship is to continue relative to alternative banks in the same municipality-year. This is the core deposit-stickiness estimate.

4.2 Determinants of deposit losses

$$Loss_{hbt} = \alpha + \beta X_{ht} + \varepsilon_{hbt}.$$

Interpretation: This regression asks who leaves money on the table by not switching to a higher-rate deposit account. The dependent variable captures the economic cost of inaction.

4.3 Depositor-borrower conversion

$$I(Borrower)_{hbt+\tau} = \alpha + \beta I(ExistingDepositor)_{hbt} + \delta_{m,a,s,e} + \varepsilon_{hbt}.$$

Interpretation: The fixed effects compare households within municipality by age, salary, and education groups. The aim is to compare depositors with otherwise similar nondepositors at the same potential bank.

4.4 Loan pricing and screening

$$LoanSpread_{hbt} = \alpha + \beta I(Depositor)_{hbt} + \gamma Controls_{hbt} + FE + \varepsilon_{hbt}.$$

$$LoanSpread_{hbt} = \alpha + \beta FutureDefault_{ht} + \gamma I(Depositor)_{hbt} + \delta FutureDefault_{ht} \times I(Depositor)_{hbt} + Controls + \varepsilon_{hbt}.$$

Interpretation: If banks use deposit histories to screen better, existing depositors should receive lower rates or the default-price gradient should be stronger for them. The paper finds the opposite pattern: a loan premium and no superior screening evidence.

4.5 Deposit pricing and pass-through

$$DepositSpread_{hbt} = \alpha + \beta PolicyRate_t + \gamma PolicyRate_t \times HighPredictedLoan_{hbt} + Controls + \varepsilon_{hbt}.$$

Interpretation: This regression asks whether relationships with higher future cross-selling value receive different deposit-price pass-through across the policy-rate cycle.

5 Identification and empirical design

- **Hypothetical relationship construction.** The paper compares actual bank-household relationships to potential relationships with other banks in the same municipality.
- **Granular fixed effects.** Municipality-year, municipality-age-salary-education, and bank-year style controls reduce the possibility that geography, lifecycle, or bank characteristics drive the results.
- **Mechanism tests.** Loan prices, default prediction, mover subsamples, tenure, and bank determinants distinguish demand-side stickiness from supply-side informational advantages.
- **Mover robustness.** If branch distance explains conversion, then moving between postcodes or municipalities should weaken the relationship. The results remain strong among movers.
- **Deposit pricing tests.** Predicted loan propensity is used to test whether banks offer better deposit terms to customers with higher future loan value.
- **Limits.** The design is observational. It cannot randomly assign deposit accounts, and some unobserved household-bank affinity may remain. The paper’s strength is the convergence of several separate tests around the same mechanism.

6 Section-by-section study guide

- **Introduction.** Read this for the three contributions: documenting cross-selling, identifying demand rather than supply mechanisms, and linking cross-selling to deposit pricing.
- **Data.** Focus on the relationship-level construction and the way hypothetical bank-household relationships are added.
- **Cross-selling of deposits.** Figure 3 and Table 2 document deposit inertia and who leaves money on the table.
- **Cross-selling of loans.** Figure 4, Tables 3–4, and Table 5 establish depositor-borrower conversion and test the loan-pricing/screening mechanism.
- **Demand mechanism.** Tables 6–7 and Figures 5–6 help separate switching costs from persistent bank preferences or branch proximity.
- **Pricing implications.** Figure 7 and Table 8 link cross-selling potential to deposit spreads and policy-rate pass-through.

7 Figures and tables

7.1 Figures 1–2: Aggregate deposit-spread environment

Read:

- Figure 1 places Norway in a broader European context by showing deposit spreads and interest-rate regimes.
- Figure 2 plots Norwegian policy rates, average deposit rates, and the difference between them over time.
- Figure 2 shows that deposit spreads are not constant: they vary across the interest-rate cycle, widening and narrowing as policy rates change.

Detailed interpretation:

- These figures are not just background. They motivate the deposit-pricing part of the paper. If spreads vary with the policy rate, then banks may change how much they are willing to subsidize deposits when future cross-selling profits are discounted differently.
- Figure 2 clarifies why the authors later study the “deposit spread beta.” A beta below one means deposit rates do not move one-for-one with policy rates. This incomplete pass-through is the object that Table 8 decomposes using market concentration and cross-selling potential.
- The figures also remind us that the value of onboarding a future borrower depends on the interest-rate environment. When rates are low, future loan profits are discounted less, making future cross-selling benefits more valuable today.

Economic message: The aggregate deposit-rate environment is the bridge between relationship-level cross-selling and monetary-policy pass-through.

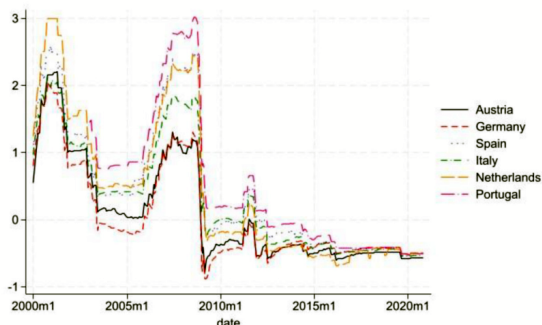


Figure 1
Deposit spreads from selected European countries
 This figure plots the European Central Bank's (ECB) Deposit Facility Rates (DFR; see <https://www.ecb.europa.eu/educational/explainers/tell-me/html/what-is-the-deposit-facility-rate.en.html>) less the average deposit rates for six selected euro area economies by month.
 Source: ECB Statistical Data Warehouse (<https://sdw.ecb.europa.eu/browse.do?node=9691394>)

(a) Figure 1: European deposit spreads

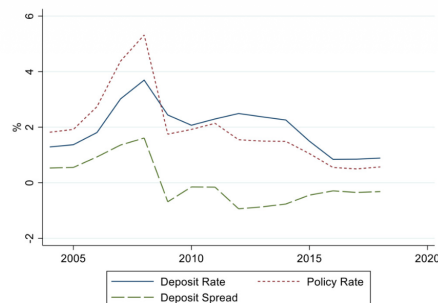


Figure 2
Average deposit spreads in Norway
 This figure plots Norwegian policy rates (<https://www.norges-bank.no/en/topics/Monetary-policy/Policy-rate/>), average deposit rates taken from www.finansportalen.no/, and their difference by year.

(b) Figure 2: Average deposit spreads in Norway

7.2 Table 1: Summary statistics

Read:

- Panel A contains 46.1 million relationship-year observations.
- The average policy rate is 1.82 percentage points; average deposit rate is 1.29; average deposit spread is 0.54.
- The average loan rate is 3.59 and average loan spread is 1.54.
- Average mean deposit volume is NOK 140,000, while final loan volume averages NOK 1.1 million.
- Average relationship duration is close to nine years.
- Households use on average 2.88 banks, which means they are not single-bank households, yet relationship inertia remains large.

Detailed interpretation:

- Table 1 gives the economic magnitude of cross-selling. A loan relationship is much larger than a deposit relationship, and loan spreads are roughly three times average deposit spreads. This is why it can be profitable to offer better deposit terms to households likely to become borrowers later.
- The deposit interest lost measure reveals the economic cost of inaction. Even in a data-rich and relatively competitive banking environment, households leave sizable deposit income on the table.
- The relationship duration statistic is central: cross-selling is not a one-period product bundle but a long relationship in which banks can onboard customers through deposits and harvest value later through loans.

Economic message: Table 1 makes the customer-lifetime-value logic plausible: future loan profits are economically large relative to current deposit margins.

Table 1
Summary statistics

	Observations	Mean	SD	Min	Max
<i>A. Relationship characteristics</i>					
Policy rate (pp)	46,132,510	1.82	1.23	0.50	5.32
Deposit rate (pp)	46,132,510	1.29	1.18	0.07	4.92
Deposit spread (pp)	46,132,510	0.54	1.37	-2.84	4.76
Loan rate (pp)	11,883,905	3.59	1.66	1.23	9.01
Loan spread (pp)	11,883,905	1.54	1.65	-3.12	7.19
Mean deposit volume (NOK)	46,132,510	140,000	300,000	0	2,000,000
Mean loan volume (NOK)	46,132,510	160,000	510,000	0	4,000,000
Final loan volume (NOK)	46,132,510	1,100,000	1,500,000	0	6,900,000
Mean household debt (NOK)	46,132,510	970,000	1,300,000	0	6,900,000
Deposit interest lost (pp)	46,132,510	2.40	1.22	-3.06	4.82
Deposit money lost (NOK)	46,132,510	320,000	480,000	0	1,800,000
Relationship duration (years)	46,132,510	8.91	4.26	1.00	15.00
<i>B. Household characteristics</i>					
I(Age < 30)	46,132,510	.34	.47	.00	1.00
I(Education group 1)	45,754,686	.15	.35	.00	1.00
I(Education group 2)	45,754,686	.42	.49	.00	1.00
I(Education group 3)	45,754,686	.30	.46	.00	1.00
I(Education group 4)	45,754,686	.13	.34	.00	1.00
I(Has real estate)	46,132,510	.53	.50	.00	1.00
House value (NOK)	46,132,510	1,200,000	1,900,000	-5,000,000	440,000,000
HH's # of banks	46,132,510	2.88	1.60	1.00	7.00
Population density	12,918,625	1.00	0.87	0.00	2.40
<i>C. Bank and market characteristics</i>					
Conversion	46,132,508	3.41	2.52	0.00	8.93
Bank deposits (NOK mio)	46,132,510	86,000	110,000	0	310,000
HHI(D)	46,132,510	.30	.12	.11	.86
HHI(L)	46,132,510	.18	.09	.08	.86
WHHI(D)	46,132,510	.30	.07	.13	.69
WHHI(L)	46,132,510	.18	.05	.00	.49
WMS(D)	46,132,510	.31	.16	.00	.69
WMS(L)	46,132,510	.16	.10	.00	.52

Panel A shows household bank relationship characteristics. Rates and spreads (against the policy rate) are expressed in percentage points (pp), volumes in Norwegian Kroners (NOK). Panel B shows household characteristics, mostly with indicators denoted as I. Panel C shows bank and municipal market characteristics. There HHI(D) and HHI(L) stand for the household municipality's Herfindahl-Hirschmann index (HHI) for the deposit (D) and loan (L) market, respectively. WHHI(D) and WHHI(L) stand for the corresponding bank-level values, weighted by the bank's portfolios across municipalities, while WMS stand for each bank's weighted market share. The conversion variable summarizes the bank-level means of each relationship's mean loan over

Table 1: Summary statistics

7.3 Figure 3 and Table 2: Deposit stickiness and deposit money lost

Read:

- Figure 3 shows that a household is more than 90 percentage points more likely to continue with an existing deposit bank after one year than to deposit at another available bank.
- The persistence declines slowly but remains economically large for many years.
- Table 2 asks which households leave more deposit interest on the table by not switching.
- Younger households, education groups, bank tenure, number of bank relationships, and other observables are used to explain deposit interest lost.

Detailed interpretation:

- Figure 3 is the deposit-market foundation of the paper. Before analyzing cross-selling into loans, the authors first show that deposit relationships themselves are persistent despite better outside rates.
- Table 2 turns persistence into a welfare-relevant object. It is one thing to show that households stay at the same bank; it is stronger to show that staying has a measurable opportunity cost.
- The key interpretation is household inaction. The results are consistent with search costs, limited attention, paperwork costs, or psychological switching costs.
- Deposit stickiness is what gives banks a relationship asset. Without persistence, deposit accounts would be weak onboarding products because customers could leave before cross-selling opportunities materialize.

Economic message: Sticky deposit relationships are the source of bank market power and the starting point for loan cross-selling.

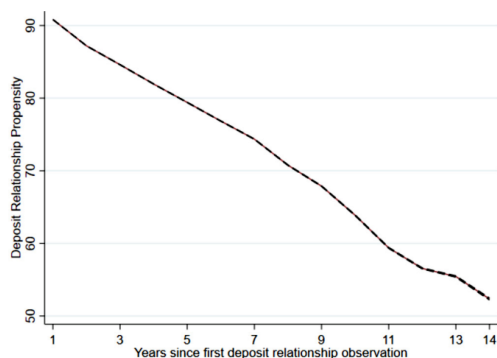


Figure 3
Depositor stickiness
Estimates are obtained using a data set where for each household $\times$ year we fill in also all other banks present at municipality and which the household could alternatively have been using. Then we control for municipality fixed effects to capture all outside options.

(a) Figure 3: Depositor stickiness

Table 2
Determinants of deposits lost by not switching banks

	(1) Deposit rate lost	(2) log deposit income lost	(3) Deposit rate lost	(4) log deposit income lost
<i>I(Parent)</i>	-0.180*** (0.001)	-0.143*** (0.002)	-0.148*** (0.001)	-0.216*** (0.002)
<i>I(Retired)</i>	-0.063*** (0.001)	0.724*** (0.001)	-0.096*** (0.001)	0.704*** (0.002)
<i>ln(Inc)</i>	0.018*** (0.000)	-0.076*** (0.001)	0.022*** (0.000)	-0.090*** (0.001)
<i>ln(Wealth)</i>	-0.047*** (0.000)	0.304*** (0.000)	-0.041*** (0.000)	0.295*** (0.000)
<i>ln(Deposits)</i>	0.004*** (0.000)	0.019*** (0.000)	0.005*** (0.000)	0.019*** (0.000)
<i>I(Age < 30)</i>	-0.264*** (0.001)	0.142*** (0.001)	-0.277*** (0.001)	0.188*** (0.001)
Constant	2.871*** (0.003)	7.908*** (0.007)	2.764*** (0.003)	8.188*** (0.008)
Observations	46,134,493	44,257,652	45,756,649	43,888,934
R^2	.019	.181	.039	.198
Fixed effects	N	N	Y	Y

The deposit rate lost for each relationship $\times$ year is the difference between the maximum rate paid by any bank in that year and municipality and the rate earned in that relationship and year. The deposit income lost is computed as the product of that rate loss and the deposit volume in each relationship and year. We then regress the rate lost and log income lost on client observables. Columns 3 and 4 include additionally fixed effects for each combination of municipality, salary quarter, and education group. Standard errors clustered by relationship are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

characteristics are plausibly correlated with both higher price sensitivity and

(b) Table 2: Determinants of deposits lost

7.4 Figure 4: Existing depositors and subsequent borrowing

Read:

- Figure 4 plots year-by-year marginal and cumulative effects of an existing deposit relationship on future borrowing from the same bank.
- The year-1 effect is roughly 4 percentage points.
- The marginal effect declines over time but remains positive and statistically significant through year 14.
- The cumulative effect is close to 20 percentage points.

Detailed interpretation:

- This is the paper's most direct visual evidence of depositor-borrower conversion. The effect is not just a short-run co-movement between deposits and loans; it persists over a long horizon.
- The declining marginal effect is intuitive: the closest-in-time loan opportunities arrive first, while later conversion events reflect slower lifecycle decisions such as home purchase after income growth or family formation.
- The cumulative curve is crucial for pricing. Banks might rationally subsidize or price deposits more aggressively because the payoff can arrive years later.
- The specification controls for granular municipality, salary, education, and age-group fixed effects and includes hypothetical bank-household relationships, so the comparison is not simply between bank customers and noncustomers in different places.

Economic message: Deposits are an onboarding product: a current deposit account creates a long-lived increased probability of a future loan relationship.

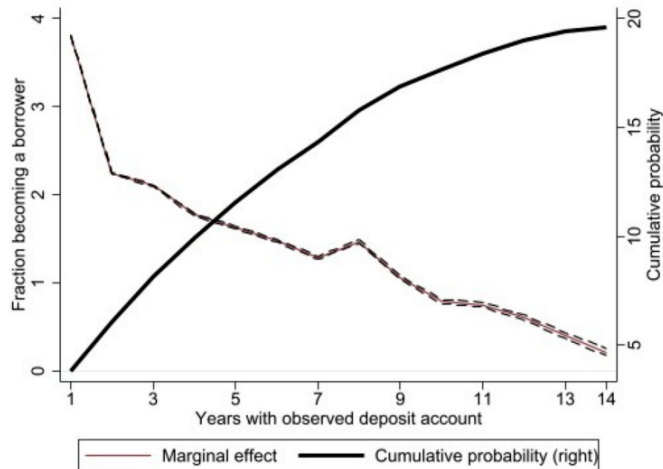


Figure 4

Existing depositors and subsequent borrowing

This figure plots our estimates of the marginal and cumulative effect of a household's opening of a deposit account with a bank on the propensity for that household to borrow from that bank in each of the subsequent years, along with 95% confidence intervals. The sample includes also hypothetical relations of each household with any other bank present in the municipality. Regressions control for municipality $\times$ salary group $\times$ education group fixed effects.

Figure 4: Existing depositors and subsequent borrowing

7.5 Tables 3–4: Loan pricing and screening

Read:

- Table 3 studies whether existing depositors receive different loan prices than nondepositors.
- Existing depositors pay a loan premium of roughly 20 basis points rather than receiving a discount.
- Table 4 studies whether banks are better at pricing future default risk for existing depositors.
- Banks charge higher loan prices to future defaulters, but the default-price relation is not stronger for existing depositors.

Detailed interpretation:

- These tables are the mechanism core of the paper. If cross-selling came from supply complementarities, banks should know existing depositors better and reward them with lower rates or screen them more accurately.
- Table 3 goes in the opposite direction: existing depositors pay more, consistent with demand-side stickiness. A depositor who already has a relationship may be less willing to search or negotiate, allowing the bank to charge a premium.
- Table 4 shows that banks price default risk, but they do not seem to have special additional information about existing depositors. This matters because mortgages are collateralized, making credit risk easier to evaluate from observable collateral and income data.
- Together, these tables shift the interpretation from relationship lending as information production to household banking as relationship lock-in.

Economic message: Cross-selling is not driven by better screening of existing depositors. It is driven by household demand-side frictions that allow banks to harvest later loan margins.

Table 3
Loan pricing

	(1) Loan rate	(2) Loan spread	(3) Loan rate	(4) Loan spread	(5) Loan rate	(6) Loan spread
<i>l</i> (Depositor)	0.147*** (0.037)	0.146*** (0.036)	0.226*** (0.027)	0.222*** (0.026)	0.267*** (0.026)	0.265*** (0.025)
<i>ln</i> (Debt)			-0.124*** (0.004)	-0.122*** (0.004)	-0.124*** (0.005)	-0.122*** (0.004)
<i>l</i> (Owns real estate)			-0.320*** (0.026)	-0.314*** (0.025)		
<i>ln</i> (Deposits)			0.001** (0.001)	0.001** (0.001)	0.001*** (0.001)	0.001*** (0.001)
Birth year			-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
Has ever received UI			0.053*** (0.002)	0.052*** (0.002)	0.045*** (0.002)	0.045*** (0.002)
Number of banks used			-0.102*** (0.005)	-0.100*** (0.005)	-0.092*** (0.005)	-0.090*** (0.005)
<i>ln</i> (House value)					-0.098*** (0.006)	-0.096*** (0.006)
Constant	3.377*** (0.021)	1.528*** (0.020)	15.212*** (0.530)	13.420*** (0.524)	16.345*** (0.597)	14.525*** (0.591)
Observations	19,990,561	19,990,561	5,196,287	5,196,287	4,601,525	4,601,525
R^2	.158	.222	.237	.297	.279	.304
Bank $\times$ Year FE	Y	Y	Y	Y	Y	Y
Municipality $\times$ Education $\times$ Year FE	N	N	Y	Y	Y	Y

Columns 1, 3, and 5 use as outcome the loan rate. Columns 2, 4, and 6 the spread thereof over the policy rate. See the main text for the rationale. "I" stands for indicators; "ln" for natural logarithms; and "UI" for unemployment insurance benefits. Standard errors clustered by bank $\times$ year are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$

(a) Table 3: Loan pricing

Table 4
Screening of future defaulters

	(1) Loan rate	(2) Loan spread
<i>Future default</i>	0.05*** (0.01)	0.05*** (0.01)
<i>l</i> (Depositor)	0.27*** (0.02)	0.26*** (0.02)
<i>Future default</i> $\times$ <i>l</i> (Depositor)	-0.02 (0.02)	-0.02 (0.02)
<i>ln</i> (Debt)	-0.12*** (0.00)	-0.12*** (0.00)
<i>ln</i> (House value)	-0.09*** (0.01)	-0.09*** (0.01)
<i>ln</i> (Deposits)	0.00*** (0.00)	0.00*** (0.00)
Birth year	-0.01*** (0.00)	-0.01*** (0.00)
Has ever received UI	0.04*** (0.00)	0.04*** (0.00)
Number of banks used	-0.09*** (0.00)	-0.09*** (0.00)
Constant	17.37*** (0.66)	15.54*** (0.66)
Observations	5,165,425	5,165,425
R^2	.27	.30
Bank $\times$ Year FE	Y	Y
Municipality $\times$ Education $\times$ Year FE	Y	Y

This table shows the results from regressing the loan rate and loan spread on a wide range of household characteristics, including a dummy for whether the household eventually defaults. Default is measured as an event where the outstanding debt is nonzero, but the bank has not received any interest payments. Standard errors clustered by bank $\times$ year are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Table 4: Screening of future defaulters

7.6 Table 5: IV regressions on depositor-borrower conversion

Read:

- Table 5 uses an IV exercise to study depositor-borrower conversion.
- The table tests whether the cross-selling result is robust to variation that shifts deposit relationships but is plausibly not directly driven by unobserved loan demand.
- The results continue to support a positive relationship between existing deposits and future borrowing.

Detailed interpretation:

- Table 5 is important because simple OLS/LPM estimates can be contaminated by unobserved household-bank match quality. For example, a household might simply like a certain bank and therefore use it for both deposits and loans.
- The IV strategy provides an additional layer of identification by isolating variation in existing deposit relationships that is less likely to reflect future loan demand directly.
- The persistence of the depositor-borrower conversion result under IV strengthens the argument that the deposit relationship itself changes later borrowing probabilities.
- The table also helps connect the paper to the customer-acquisition logic: a bank may want to obtain a deposit relationship even before a household is a borrower because that relationship has predictive power for later lending.

Economic message: The conversion effect is not only a raw correlation between deposit and loan use. It survives a stronger identification exercise.

Table 5
Instrumental variable regressions on depositor-borrower conversion

	(1) <i>I(Future loan)</i>	(2) <i>I(Future loan)</i>	(3) <i>I(Depositor)</i>	(4) <i>I(Future loan)</i>	(5) <i>I(Future loan)</i>	(6) <i>I(Future loan)</i>	(7) <i>I(Depositor)</i>	(8) <i>I(Future loan)</i>
<i>I(Existing depositor)</i>	0.080*** (0.000)			0.205*** (0.000)	0.030*** (0.000)			0.110*** (0.001)
<i>Old dep market share</i>		0.244*** (0.000)	1.189*** (0.000)			0.019*** (0.000)	0.176*** (0.000)	
<i>Current dep market share</i>					0.205*** (0.000)	0.326*** (0.000)	1.029*** (0.000)	0.114*** (0.001)
Constant	0.002*** (0.000)	0.001*** (0.000)	0.008*** (0.000)		0.001*** (0.000)	0.001*** (0.000)	0.006*** (0.000)	
Observations	110,640,774	110,640,774	110,640,774	110,640,774	110,640,774	110,640,774	110,640,774	110,640,774
<i>R</i> ²	.067	.080	.223	-.052	.102	.099	.294	.047
Regression Cluster	OLS M ⁴ A ³ S ² E	RF M ⁴ A ³ S ² E	FS M ⁴ A ³ S ² E	IV M ⁴ A ³ S ² E	OLS M ⁴ A ³ S ² E	RF M ⁴ A ³ S ² E	FS M ⁴ A ³ S ² E	IV M ⁴ A ³ S ² E

This table restricts attention to a subsample of movers and shows the results from an IV regression where *I(Existing depositor)* is instrumented using the market share of the bank in the municipality. The sample is filled in to include for each household also potential relationships with all banks doing business in that household's current municipality, as explained in main text. "RF" corresponds to reduced form, that is, regressing the key dependent variable directly on the instrument, while "FS" corresponds to the first stage. All six columns contain

Table 5: IV regressions on depositor-borrower conversion

7.7 Tables 6–7 and Figures 5–6: Switching costs, account types, movers, and household determinants

Read:

- Table 6 shows that the probability of switching banks when taking a new loan falls with bank tenure.
- Table 7 relates conversion probabilities to household and bank characteristics; older, wealthier, and less multi-banked households are more likely to convert.
- Figure 5 shows deposit spreads by account type, illustrating how banks segment deposit pricing across eligibility groups.
- Figure 6 shows that about half of new loan relationships begin after the household has moved to a different postcode or municipality since starting the deposit relationship.

Detailed interpretation:

- Table 6 is a switching-cost test. If the relationship effect were just persistent tastes for a bank, tenure should not necessarily reduce switching when a household takes out a loan. The negative tenure effect suggests that the relationship itself accumulates inertia.
- Table 7 gives a richer picture of who is most cross-sellable. Households with fewer bank relationships, older households, and wealthier households are easier targets for relationship conversion. These characteristics line up with search and information-gathering costs rather than hidden bank-specific preference alone.
- Figure 5 is important because it shows explicit deposit price discrimination by product type. Young accounts, student accounts, minimum-volume accounts, and retirement accounts have different spreads. This supports the paper's argument that banks can price deposits strategically at onboarding.
- Figure 6 addresses the branch-distance alternative. If conversion were just driven by proximity to the original branch, moving across postcodes or municipalities should break the link. The large share of conversions after moves weakens the pure-location story.

Economic message: The demand-side mechanism is not just “customers like the bank.” It is accumulated switching cost, segmented deposit pricing, and household inertia.

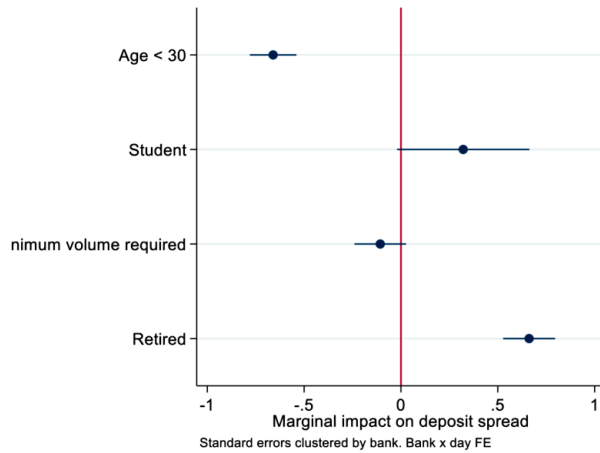
Table 6
Switching as a function of bank tenure

	(1) Switcher	(2) Switcher
$I(\text{Age} < 30)$	0.069*** (0.000)	
$I(\text{Wealth} < P50)$	0.116*** (0.000)	0.052*** (0.002)
$I(\text{University})$	0.021*** (0.000)	
Bank tenure	-0.055*** (0.000)	-0.050*** (0.001)
Constant	0.395*** (0.001)	0.449*** (0.003)
Observations	3,307,023	3,258,934
R^2	.298	.357
Municipality $\times$ Age $\times$ Salary $\times$ Education FE	N	Y

The sample investigates for all household $\times$ year combinations in which a new loan is taken out whether the household did already hold deposits at that bank before (*Switcher* = 0; about 75% of cases) or not (*Switcher* = 1 about 25% of cases). Standard errors are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

condition on a sample of movers and (b) instrument the existing deposits

(a) Table 6: Switching and bank tenure



5
t spreads by account type
ients from regressing deposit spreads on different account types, based on data from Finansportalen.

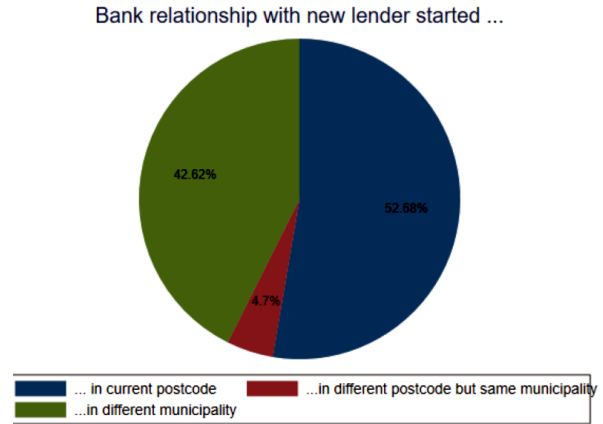
(a) Figure 5: Deposit spreads by account type

Table 7
Household and bank determinants of loan probability, loan amount, and deposit pricing

	(1) $I(\text{Loan})$	(2) $\ln(\text{FinalDebt})$	(3) Dspread	(4) $I(\text{Loan})$	(5) $\ln(\text{FinalDebt})$	(6) Dspread
$I(\text{Parent})$	-0.022*** (0.000)	0.635*** (0.005)	-0.156*** (0.003)	-0.023*** (0.000)	0.446*** (0.005)	-0.135*** (0.009)
$I(\text{Retired})$	0.013*** (0.000)	-2.797*** (0.003)	-0.040*** (0.001)	0.029*** (0.000)	-1.235*** (0.004)	-0.029*** (0.004)
$\ln(\text{Inc})$	0.008*** (0.000)	0.932*** (0.002)	-0.011*** (0.000)	0.003*** (0.000)	0.590*** (0.002)	-0.016*** (0.002)
$\ln(\text{Wealth})$	0.001*** (0.000)	0.142*** (0.001)	-0.027*** (0.000)	0.001*** (0.000)	0.086*** (0.001)	-0.027*** (0.002)
$\ln(\text{Deposits})$	0.003*** (0.000)	-0.062*** (0.000)	-0.005*** (0.000)	0.003*** (0.000)	-0.059*** (0.000)	-0.005*** (0.001)
$I(\text{Age} < 30)$	0.075*** (0.000)	0.999*** (0.003)	-0.258*** (0.001)	0.074*** (0.000)	0.827*** (0.003)	-0.281*** (0.007)
$HH's \# \text{ of banks}$	-0.013*** (0.000)	0.582*** (0.001)	-0.086*** (0.000)	-0.014*** (0.000)	0.478*** (0.001)	-0.078*** (0.002)
Multi-product HH			0.086*** (0.001)			0.082*** (0.010)
Constant	-0.027*** (0.001)	-3.759*** (0.020)	1.608*** (0.003)	-0.007*** (0.001)	1.229*** (0.025)	1.644*** (0.013)
Observations	17,090,102	17,090,102	7,806,169	16,860,604	16,860,604	7,684,772
R^2	.047	.199	.027	.058	.229	.041
M* S^2 FE	N	N	N	Y	Y	Y
Cluster	Rel	Rel	Rel	M* S^2 E	M* S^2 E	M* S^2 E

Household characteristics until the indicator for age below 30 as above. $HH's \# \text{ of banks}$ is the number of banks with which the household entertains a relationship in that year. *Multi-product HH* is an indicator for households that have both deposits and a loan in that bank relationship. Columns 1–3 without fixed effects and with standard errors clustered by bank relationship. Columns 4–6 include municipality $\times$ salary $\times$ education fixed effects and cluster standard errors by these same categories. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Table 7: Household and bank determinants



re 6
tion of bank relationships started before a move
figure considers for all new debt a household owes to a bank which fraction of households have already moved to a different postcode area within or across municipalities since starting to deposit at that bank.

(b) Figure 6: Relationships before a move

7.8 Figure 7 and Table 8: Deposit-spread beta and predicted loan propensity

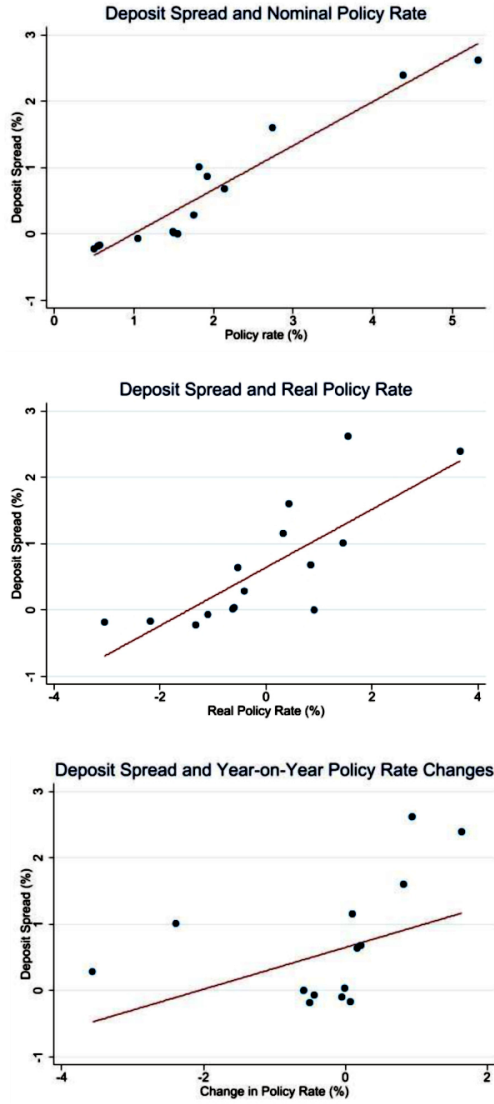
Read:

- Figure 7 plots deposit spreads against nominal policy rates, real policy rates, and changes in nominal policy rates.
- The top panel shows a positive relationship between policy rates and deposit spreads; deposit rates move less than one-for-one with policy rates.
- Table 8 estimates a baseline deposit spread beta of about 0.67.
- Table 8 interacts the policy rate with market concentration and with above-median predicted loan propensity.
- The interaction between policy rate and predicted loan propensity is positive and significant across specifications.

Detailed interpretation:

- Figure 7 establishes the pass-through fact at the relationship level. As policy rates rise, deposit spreads tend to rise, meaning deposit rates do not fully keep up with the policy rate.
- Table 8 asks whether this incomplete pass-through reflects only market power or also cross-selling. The table includes several measures of concentration and market share, and still finds that high predicted loan propensity increases the deposit spread beta.
- The interpretation is subtle. A higher beta for high-predicted-loan customers means their deposit rates are less sensitive to policy rates. At low rates, future cross-selling profits have higher present value, so banks are more willing to offer attractive deposits to these customers. As rates rise, the present value of future loan profits falls and deposit spreads widen more.
- This result connects micro cross-selling to macro monetary transmission: relationship value can change pass-through even for deposits, not just loans.

Economic message: Deposit pricing embeds customer lifetime value. Banks price deposits differently when the customer is more likely to become a future borrower, and this affects deposit-rate pass-through.



(a) Figure 7: Deposit spread beta

Table 8
The deposit spread beta by predicted loan propensity and market concentration

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	<i>Dspread</i>	<i>Dspread</i>	<i>Dspread</i>	<i>Dspread</i>	<i>Dspread</i>	<i>Dspread</i>	<i>Dspread</i>
<i>Policy rate (PR)</i>	0.671*** (0.000)	0.662*** (0.000)	0.649*** (0.000)	0.649*** (0.000)	0.605*** (0.000)	0.670*** (0.000)	0.589*** (0.000)
<i>I(Conc > P50)</i>		-0.052*** (0.001)	-0.065*** (0.001)	-0.235*** (0.001)	-0.142*** (0.001)	0.080*** (0.001)	0.066*** (0.001)
<i>PR*I(Conc > P50)</i>		-0.011*** (0.000)	0.016*** (0.000)	0.023*** (0.000)	0.096*** (0.000)	-0.020*** (0.000)	0.063*** (0.000)
<i>I(PredLoan > P50)</i>		-0.089*** (0.001)	-0.091*** (0.001)	-0.109*** (0.001)	-0.095*** (0.001)	-0.072*** (0.001)	-0.081*** (0.001)
<i>PR*I(PredLoan > P50)</i>		0.045*** (0.000)	0.049*** (0.000)	0.053*** (0.000)	0.054*** (0.000)	0.044*** (0.000)	0.054*** (0.000)
Constant	-0.689*** (0.000)	-0.630*** (0.001)	-0.629*** (0.001)	-0.530*** (0.001)	-0.605*** (0.001)	-0.720*** (0.001)	-0.662*** (0.001)
Observations	45,756,653	45,756,653	45,756,653	45,756,653	45,756,653	45,756,653	45,756,653
<i>R</i> ²	.383	.384	.383	.388	.385	.383	.387
MC measure	None	HHI(D)	HHI(L)	WHHI(D)	WHHI(L)	WMS(D)	WMS(L)

Dspread is the deposit spread between the policy rate and the deposit rate by bank household relationship and year. We regress it first on the Norwegian central bank's policy rate (PR) to approximate changes in the discount rates to be applied to future cross-selling profits. The next regressor is *I(Conc > P50)*, an indicator for market concentration above the sample median. Market concentration (MC) is measured as the municipal deposit market Herfindahl-Hirschmann index (HHI) in column 2, the municipal loan market HHI in column 3, the bank-level weighted deposit (loan) market HHI in column 4 (5), and the bank-level-weighted deposit (loan) market share in column 6 (7). The other indicator of interest is *I(PredLoan > P50)*, which denotes bank-household relationship for whom borrowing propensities predicted by the regressions in Table 4 are above the median. After the PR and the two indicators just described come the regressors of most interest in this table, the *interactions* between the PR and the two indicators, respectively. All columns control for age group $\times$ salary group $\times$ education group fixed effects. Standard errors clustered by relationship are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Table 8: Deposit spread beta

8 Result map

- **Deposit inaction:** households remain with deposit banks even when other banks offer better rates.
- **Loan conversion:** a deposit relationship raises the probability of future borrowing from the same bank by close to 20 percentage points over 14 years.
- **Loan pricing:** existing depositors pay a loan premium, which supports demand-side stickiness rather than supply-side information discounts.
- **Screening:** banks price future default risk but not more precisely for existing depositors.
- **Mechanism:** tenure, mover, and household-characteristic tests point toward switching costs and inaction.
- **Deposit pricing:** predicted future loan value is incorporated into deposit terms.
- **Pass-through:** cross-selling potential helps explain deposit-spread beta heterogeneity.

9 Short summary

- The paper studies cross-selling in household banking using Norwegian administrative relationship-level data.
- Existing depositors are extremely likely to stay with the same bank, even when they could earn higher deposit rates elsewhere.
- Existing depositors are also much more likely to become future borrowers from the same bank.
- Loan pricing and default tests reject the idea that banks cross-sell mainly because they have superior information about depositors.
- The evidence points to demand complementarities and switching costs.
- Banks internalize future cross-selling profits in deposit pricing.
- Cross-selling potential affects deposit-rate pass-through, connecting relationship banking to monetary transmission.

10 Conclusion

10.1 Core conclusion

Basten and Juelsrud show that bank-household relationships are multiproduct relationships. A deposit account is not only a funding source for the bank; it is an acquisition channel for future loan business.

10.2 Economic intuition

Households often begin bank relationships with deposits and only later need mortgage or other loan products. Because deposit relationships are sticky, banks can use deposits to create future lending opportunities. The key point is that the future value of a depositor includes expected future loan profits.

10.3 Mechanism conclusion

The evidence supports demand complementarities over supply complementarities. Existing depositors pay a loan premium, banks do not screen them better, and mover/tenure tests point toward switching costs rather than persistent bank preferences or branch proximity.

10.4 Pricing conclusion

Banks price deposits with future cross-selling in mind. Customers predicted to become borrowers receive better deposit terms, and their deposit spreads respond differently to policy rates.

10.5 Policy implications

The paper matters for deposit competition, market power, financial consumer protection, and monetary-policy pass-through. If banks value deposits partly because of future loan sales, deposit pricing can reflect customer lifetime value rather than only current funding costs.

10.6 Limitations

The paper is observational, and some unobserved household-bank affinity may remain. It also studies Norway, where mortgage lending is collateralized and banks have rich credit information. The supply-versus-demand balance may differ in unsecured consumer credit or small business lending.

10.7 Takeaway

The key takeaway is that household banking relationships have option value. Cross-selling turns deposits into a long-lived customer-acquisition product, and this changes how banks price both deposits and loans.